

CariSurg MedTech Pathways Research Project

Preliminary Proposal

Project Title:

Explainable AI-Assisted Emergency Department Triage Support System for Resource-Constrained Healthcare Settings

Student Name:

Azaria DaSilva

Statement of the Problem

Emergency Department (ED) triage aims to prioritize patients according to urgency, but decisions occur across multiple workflow stages where information becomes available gradually rather than simultaneously. Registration, vital sign collection, nurse assessment, and queue management each introduce opportunities for delay, incomplete information, and communication loss. As a result, prioritization decisions may occur before the full clinical picture is available. These workflow realities can reduce consistency and increase operational pressure. Therefore, any AI-assisted triage solution must improve decision support while fitting naturally into existing clinical processes.

Proposed Solution

This project proposes an AI-assisted clinical decision support system integrated into the Emergency Department triage workflow. Rather than replacing nurse judgement, the system would analyze information as it becomes available throughout registration, vital sign collection, and triage assessment to provide contextual support. The proposed system would identify abnormal observations, surface risk indicators, and support prioritization decisions while preserving clinician autonomy. It would operate passively alongside existing processes to minimize additional documentation and workflow disruption. By continuously updating recommendations as new information becomes available, the system aims to support timely assessment while respecting operational realities and maintaining human oversight.

Literary Review

[1] This study addressed the problem of emergency department overcrowding and inconsistent triage decision-making. The authors used machine learning and natural language processing techniques to analyze patient data and support triage prioritization and resource allocation. The approach demonstrated improved efficiency, including reduced waiting times and improved time-to-treatment and bed allocation. However, the study was limited by concerns around algorithmic bias, lack of transparency in model decision-making, and clinician distrust of black-box systems.

[2] This review addressed the challenge of improving patient classification accuracy in emergency department triage systems. The authors used a scoping review methodology to analyze existing artificial intelligence applications trained on demographic, symptom, and clinical datasets. The findings showed that AI systems often outperformed traditional triage methods in predictive accuracy and prioritization. However, limitations included reliance on retrospective datasets, inconsistent preprocessing techniques, and insufficient external validation across healthcare settings.

[3] This study addressed the problem of improving triage accuracy in high-volume emergency departments. The authors used a stacked ensemble machine learning approach combining six supervised learning models trained on historical emergency department data. The system achieved an accuracy of 80.05% and an AUROC of 0.93, demonstrating strong predictive performance. However, the study was limited by its use of retrospective data from only two hospitals in Korea, restricting generalizability.

[4] This study addressed the challenge of improving triage decision-making using both structured and unstructured clinical data. The authors used machine learning and natural language processing to analyze triage records and free-text clinical notes for outcome prediction. The model showed improved predictive performance compared to conventional triage approaches. However, limitations included poor generalizability and dependence on local documentation styles and hospital-specific data structures.

[5] This study addressed the problem of evaluating whether large language models can support emergency department triage decision-making. The authors used generative AI models to simulate triage decisions across clinical scenarios and compared their performance with inexperienced human decision-makers. The results showed promising performance of large language models in triage classification tasks. However, concerns remained regarding reliability, lack of transparency, and safety risks in real-world clinical deployment.

[6] This study addressed the problem of predicting 72-hour emergency department return visit admissions to identify high-risk patients early. The authors used multiple machine learning algorithms, including logistic regression and gradient boosting, trained on electronic health record data from multiple hospitals. The model achieved strong predictive performance with an AUC up to 0.87 in development and 0.75 in external validation. However, limitations included reduced generalizability, dataset imbalance, and limited clinical actionability in real-world workflows.

[7] This study addressed the challenge of improving interpretability in machine learning-based emergency department triage systems. The authors used an interpretable machine learning framework designed to produce transparent prioritization scores and address class

imbalance in clinical datasets. The model improved triage performance while maintaining explainability for clinicians. However, limitations included dependency on local population characteristics and uncertain transferability to other healthcare environments.

[8] This study addressed inefficiencies in patient flow within a Caribbean emergency department. The authors used qualitative workflow analysis to examine registration, communication, and clinical coordination processes. The findings identified bottlenecks caused by information delays, workload pressures, and communication breakdowns. However, the study was limited by its qualitative design and lack of quantitative performance metrics, limiting generalizability.

[9] This study addressed the problem of improving triage accuracy and outcome prediction compared to traditional methods. The authors used machine learning models trained on electronic triage data to predict clinical outcomes and prioritization levels. The results showed that machine learning improved differentiation of high-risk patients compared to conventional triage systems. However, limitations included challenges in integration into clinical workflows and the need for clinician oversight in decision-making.

[10] This study addressed the challenge of implementing clinical decision support systems effectively in healthcare environments. The authors used an evaluative approach examining adoption factors such as usability, workflow integration, and clinician training. The findings showed that successful implementation depends heavily on system design, trust, and alignment with clinical workflows. However, limitations included variability across healthcare settings and the difficulty of standardizing adoption conditions.

Healthcare Workflow

Patient Arrival



Registration

(Administrative details collected; patient reason for visit recorded)



Vitals Collection

(Vital signs measured and temporarily held until triage assessment)



Triage Nurse Assessment

(Chief complaint + clinical judgement + urgency assignment)



Priority Assignment

(Patient enters queue based on assigned urgency)



Waiting Area Monitoring

(Patient condition may change while waiting)



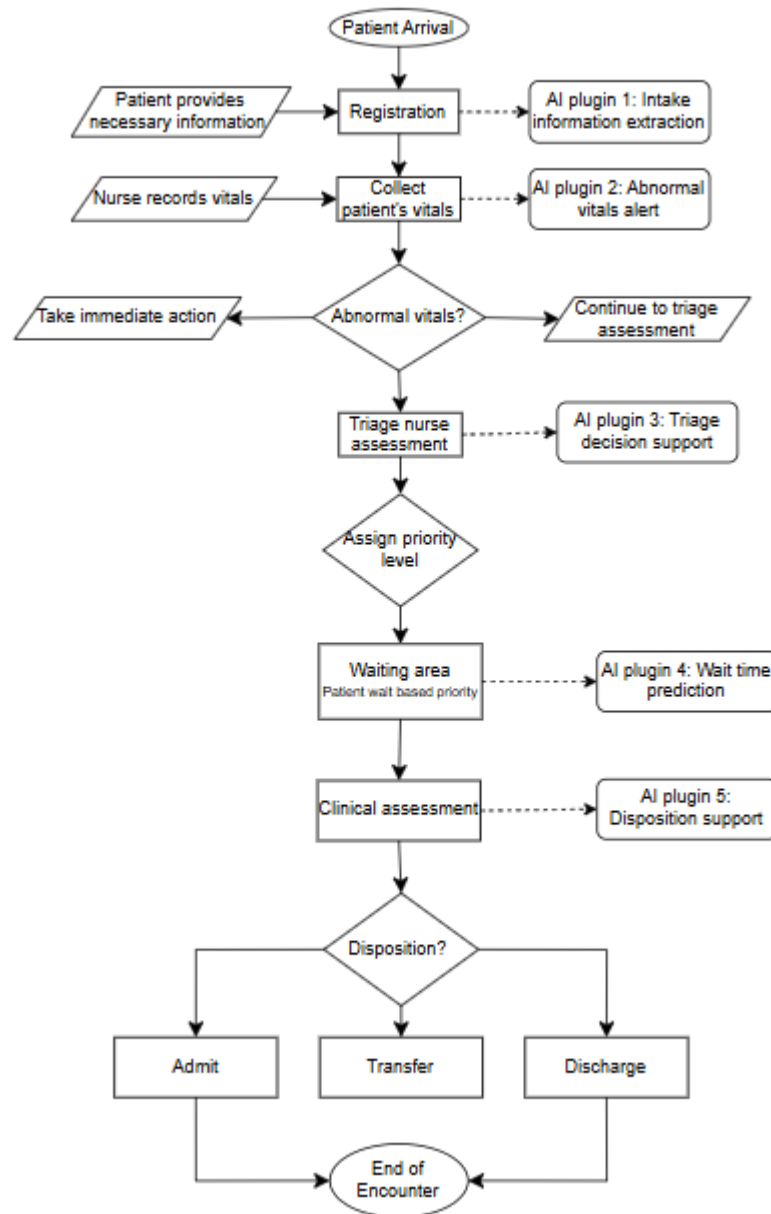
Clinician Assessment



Disposition

(Discharge / Admit / Transfer)

Information enters progressively throughout the workflow and clinical decisions are revised as additional information becomes available.



AI Integration Points

AI Plug-in 1 - Registration Support

Purpose: Structure intake information into standard fields before assessment begins.

AI Plug-in 2 - Observation Validation

Purpose: Detect missing or unusual combinations of vital signs before triage completion.

AI Plug-in 3 - Triage Decision Support

Purpose: Present risk indicators alongside nurse assessment rather than replacing categorization.

AI Plug-in 4 - Queue Monitoring

Purpose: Identify patients whose wait duration and observations suggest reassessment.

AI Plug-in 5 - Disposition Awareness

Purpose: Support clinicians by surfacing relevant outcome indicators.

Constraints

Constraint 1: Sequential Information Capture Creates an Early Decision Blind Spot

Workflow observation:

Registration occurs before full observations and assessment are available.

Mechanism:

Initial prioritization decisions may begin before complete vital signs and clinical context are captured.

AI implication:

The system must update recommendations dynamically rather than assuming complete data availability.

Design response:

Display confidence levels and continuously refresh outputs.

Measurement:

Percentage of recommendation changes after vital completion.

Constraint 2: Multiple Handoffs Create Information Translation Risk

Workflow observation:

Patient information moves across registration staff, triage nurses, waiting stages, and clinicians.

Mechanism:

Information gathered early may not remain visible or equally emphasized later.

AI implication:

The system must preserve context across transitions.

Design response:

Carry forward structured summaries instead of requesting repeated entry.

Measurement:

Number of repeated data collection events per patient.

Constraint 3: Queue Status Does Not Automatically Trigger Reassessment

Workflow observation:

Patients wait according to assigned urgency while clinical condition may evolve.

Mechanism:

A patient may deteriorate after triage without triggering reassessment.

AI implication:

AI should monitor elapsed waiting time together with available observations.

Design response:

Generate reassessment prompts rather than reassign urgency automatically.

Measurement:

Time between deterioration indicators and reassessment.

Stakeholders

Triage Nurse

Concern: Maintaining assessment speed without duplicate documentation.

Emergency Physician

Concern: Receiving reliable prioritization support.

Patient

Concern: Fair and timely assessment.

Hospital Administration

Concern: Improving patient flow without increasing workload.

Clinical IT Team

Concern: Integration into existing infrastructure.

Reflection

The initial proposal focused primarily on improving triage prediction. Workflow mapping shifted the focus toward understanding how information moves through the Emergency Department and where delays emerge. This highlighted that healthcare AI creates value only when it adapts to operational realities, preserves clinician judgement, and fits existing patterns of work. Model performance alone is insufficient if recommendations arrive at the wrong stage of the workflow or increase documentation burden.

References

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